# Double Number Line Ratio Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Double Number Line Ratio Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

## 1 Predict

before you touch the tool

1. A recipe uses 3 cups of juice concentrate for every 5 cups of water. Will 5 cups of concentrate with 7 cups of water taste the same? Explain.

   No. Adding 2 to both parts changes the ratio: 3 : 5 is 0.6 cup of concentrate per cup of water, but 5 : 7 is about 0.71, so it is stronger.

2. Guess: how much water goes with exactly 1 cup of concentrate?

   $\frac{5}{3} = 1\frac{2}{3}$ cups of water.

## 2 Explore

use the tool

1. Slide the marker to 9 cups of concentrate. How much water goes with it? What number multiplied both amounts?

   15 cups of water. Both amounts were multiplied by 3.

2. Move the marker so the top line reads exactly 1. What does the bottom line show? What is this number called?

   $\frac{5}{3}$ cups of water: the unit rate (water per cup of concentrate).

3. Now find how much concentrate goes with exactly 1 cup of water.

   $\frac{3}{5}$ cup of concentrate.

4. Type 5 and 7 into Test a pair. What does the tool say, and where is the point on the graph?

   Not equivalent: the top is multiplied by $\frac{5}{3}$ but the bottom by $\frac{7}{5}$. The point is off the line.

## 3 Record

use the tool and fill in the table

### Juice recipe, 3 : 5

Use the marker. The last column is "multiplied by" compared with 3 : 5.

| Concentrate (cups) | Water (cups) | Water per cup of concentrate | Multiplied by |
|---|---|---|---|
| 3 | 5 | $\frac{5}{3}$ | × 1 |
| 6 | 10 | $\frac{5}{3}$ | × 2 |
| 12 | 20 | $\frac{5}{3}$ | × 4 |
| $\frac{3}{2}$ | $\frac{5}{2}$ | $\frac{5}{3}$ | × $\frac{1}{2}$ |
| $\frac{3}{5}$ | 1 | $\frac{5}{3}$ | × $\frac{1}{5}$ |

### Road trip, 2 hours : 7 miles

Choose Road trip.

| Hours | 2 | 4 | ?1 | ?6 |
|---|---|---|---|---|
| Miles | 7 | 14 | $3\frac{1}{2}$ | 21 |

## 4 Explain what did you discover?

1. Equivalent ratios come from multiplying (or dividing) both amounts by the same number.
2. The unit rate is the amount that goes with exactly 1. For equivalent ratios it is always the same.
3. On the graph, equivalent pairs lie on one straight line through (0, 0).
4. A classmate says 4 : 6 and 6 : 8 are equivalent because you add 2 to each part. Explain why that is wrong. Use the unit rate or the double number line.
   4 : 6 is 1.5 per 1 (6 ÷ 4), but 6 : 8 is about 1.33 per 1 (8 ÷ 6). Adding 2 does not multiply both parts by the same number, so the ratio changes.

## 5 Apply a new situation, no tool

1. Apples cost $6 for 4 pounds. Find the price per pound and the cost of 10 pounds.
   $1.50 per pound; 10 pounds cost $15.
2. Which is the better buy: 3 notebooks for $4.50 or 5 notebooks for $7.00?
   $1.50 each vs. $1.40 each: 5 for $7.00 is the better buy.
3. Fill in the blanks: 2 : 7 = 8 : ___ = ___ : 35
   28 and 10
4. A car travels 150 miles in $2\frac{1}{2}$ hours. At the same rate, how far does it go in 4 hours?
   60 miles per hour, so 240 miles.

### Teacher notes

**Intended discovery:** Equivalent ratios come from multiplying both quantities by the same factor, so the unit rate is constant and every equivalent pair sits on one line through the origin.

**Likely misconceptions:**

- Additive thinking: "add the same amount to both parts" (Explore 4 and Explain target this).
- Mixing up the two unit rates ($\frac{5}{3}$ cups of water per cup of concentrate vs. $\frac{3}{5}$ cup of concentrate per cup of water).
- Believing ratios only use whole numbers; the marker shows $\frac{3}{2} : \frac{5}{2}$ is the same ratio.

**Answers that can vary:** Test-a-pair choices in Explore 4 and in the Try this list; any pair (3k, 5k) is correct.